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POSITIONS IN CSS

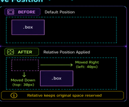
Default position is static

Relative Position

CODE EXAMPLE

```
.parent{
  width: 400px;
  height: 220px;
  border: 2px solid cyan;
}

.box{
  position: relative;
  top: 30px;
  left: 40px;
}
```



Relative keeps original space reserved

HOW RELATIVE POSITIONING WORKS

position: relative → Element moves visually → Original space remains reserved → Layout does not collapse

01 MOVES FROM ORIGINAL POSITION
The element is visually moved from where it would normally appear.

02 SPACE REMAINS RESERVED
The space of the element is kept in the normal flow of the document.

03 USEFUL FOR SMALL ADJUSTMENTS
Ideal for fine-tuning position without affecting other layout elements.

04 WORKS WITH TOP, LEFT, RIGHT, BOTTOM
Use top, left, right, or bottom to control the direction of movement.

2) ABSOLUTE POSITION

WHAT IS ABSOLUTE POSITION?

Absolute positioning allows you to position an element relative to its nearest positioned ancestor.

KEY IDEA

After the element moves, its original position can be occupied by other elements.

IMPORTANT POINTS

- Its original position can be occupied by other elements.
- It is positioned relative to the nearest positioned ancestor (an element with position other than static).
- If no positioned ancestor is found, it is positioned relative to the initial containing block (viewport).
- It can overlap other elements.
- Useful for overlays, dropdowns, tooltips, icons, badges, etc.

HOW ABSOLUTE POSITION WORKS

01 **Positioned Relative to Nearest Positioned Ancestor**
It looks for its nearest ancestor that has a position value other than static (i.e., relative, absolute, fixed, or sticky).

02 **If No Positioned Ancestor is Found**
It is positioned relative to the initial containing block, which is usually the viewport.

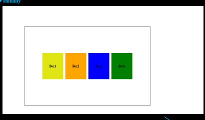
03 **Position Controlled by Offsets**
You can use top, right, bottom, and left to move the element from the reference edges of its ancestor (or viewport).

04 **Can Overlap Other Elements**
Since it does not take space in the normal flow, it can appear over other elements in the layout.

WHEN TO USE ABSOLUTE POSITION?

- For precise placement inside a container.
- For overlays and elements that should appear above others.
- For custom dropdowns, tooltips, popups, badges, etc.
- For positioning elements independent of normal document flow.

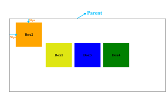
Initially



Now let's give css

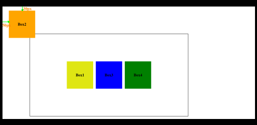
```
.parent{ position:relative;}

.box2{
        background-color: orange;
        position: absolute;
        top: 30px;
        left: 50px;
}
```



If no position is assigned to any parent element, the absolutely positioned element will move relative to the viewport.

```
.box2{
        background-color: orange;
        position: absolute;
        top: 30px;
        left: 50px;
}
```



3) FIXED POSITION

CODE EXAMPLE

```
.box-1 {
  position: fixed;
  top: 0;
  left: 20px;
}
```

WHAT IS FIXED POSITION?

It is similar to absolute position, but the only difference is that it is relative to the viewport and not to any parent element.

NOTE:
When the position of any element is fixed, no matter how we scroll the page, that element does not move.

KEY POINTS

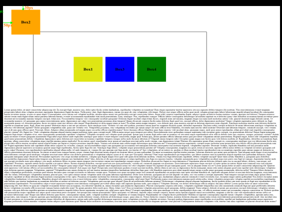
- Same as absolute, but it is relative to the viewport.
- It is not relative to any parent element.
- After fixing an element, its position on the screen remains the same even after scrolling.
- It is attached to the viewport (top, bottom, left, right).
- Commonly used for navigation bars, headers, footers, side menus, etc.

COMMON USES OF FIXED POSITION

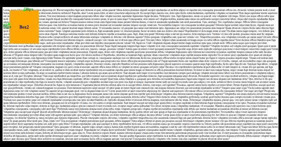
- Navigation bars that stay on top
- Footers that stay at bottom
- Side menus that remain visible
- Notification bars or alerts

Without Scrolling

```
.box2{
        background-color: orange;
        position: fixed;
        top: 30px;
        left: 50px;
}
```



After Scrolling



4) STICKY POSITION

WHAT IS STICKY POSITION?

It allows an element to stick to its nearest scrolling parent]based on the specified offset (top, bottom, left, or right).
Initially, it behaves like **simple element**

KEY IDEA

An element will act as relative until it reaches the specified offset, then it will "stick" and behave like fixed within its scrolling parent.

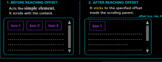
HOW STICKY POSITION WORKS

- It sticks to its nearest scrolling parent, not the viewport.
- The element sticks when the scroll position crosses the specified offset (top, bottom, left, or right).
- It will stay stuck until the end of its scrolling parent.
- When the scrolling parent ends, the element stops sticking.

HOW IT WORKS (VISUAL FLOW)

1. BEFORE REACHING OFFSET
Acts like **simple element.**
It scrolls with the content.

2. AFTER REACHING OFFSET
It sticks to the specified offset inside the scrolling parent.



- box-1, box-2, and box-3 are inside a scrolling parent element.
- At first, all boxes scroll with the content (normal position).
- When the scroll reaches the offset (e.g., top: 20px), the **box** stick at that position.
- **This** remain stuck until the end of the scrolling parent.
- After that, **this** stop sticking.

COMMON USES

- Sticky navigation bars
- Table headers
- Sidebar menus
- Call-to-action bars or banners